

Q4(a)	The expected value of the current through a resistor is 20 mA. However the measurement yields a current value of 18 mA. Calculate (i) absolute error (ii) % error and (iii) relative accuracy	3	CO4	K5																						
Q4(b)	The below table gives the set of ten measurement that were recorded in the laboratory. Calculate the precision of 5th measurement. <table><tr><th>Measurement Number</th><th>Measurement Value</th></tr><tr><td>1.</td><td>98</td></tr><tr><td>2.</td><td>101</td></tr><tr><td>3.</td><td>102</td></tr><tr><td>4.</td><td>97</td></tr><tr><td>5.</td><td>101</td></tr><tr><td>6.</td><td>100</td></tr><tr><td>7.</td><td>103</td></tr><tr><td>8.</td><td>98</td></tr><tr><td>9.</td><td>106</td></tr><tr><td>10.</td><td>99</td></tr></table>	Measurement Number	Measurement Value	1.	98	2.	101	3.	102	4.	97	5.	101	6.	100	7.	103	8.	98	9.	106	10.	99	2	CO4	K5
Measurement Number	Measurement Value																									
1.	98																									
2.	101																									
3.	102																									
4.	97																									
5.	101																									
6.	100																									
7.	103																									
8.	98																									
9.	106																									
10.	99																									
Q4(c)	Explain the construction and working of LVDT.	5	CO4	K2																						

4(a)	Discuss the principle of operation of a Linear Variable Differential Transformer (LVDT). Also, state its advantages and disadvantages.	5	CO4	K2, K4
4(b)	What is transducer? Explain the difference between active and passive transducer with examples.	5	CO4	K1,

4(a)	Discuss the principle of operation of a Linear Variable Differential Transformer (LVDT). Also, state its advantages and disadvantages.	5	CO4	K2, K4
4(b)	What is transducer? Explain the difference between active and passive transducer with examples.	5	CO4	K1, K2

4.(a)	Show the working of LVDT with proper diagrams.	5	4
(b)	Expected value of voltage across a resistor is 90 V. However measured value is 88.5 V. Calculate (i) Percentage Error (ii) Relative Accuracy	2	4
(c)	Explain the functional block diagram of an instrument.	3	4